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REVIEW ARTICLE

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# What is habit and how can it be used to change real-world behaviour? Narrowing the theory-reality gap

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## Abstract

Habit change is often seen as key to successful long-term behaviour change. Making 'good' behaviours habitual—that is, ensuring a behaviour is prompted automatically on exposure to situational cues, based on cue-response associations learnt through context-consistent repetition—is portrayed as a mechanism for sustaining such behaviours over time. Conversely, disrupting 'bad' habits is expected to terminate longstanding unwanted actions. Yet, some commentators have suggested that the role of habit in real-world behaviour and behaviour change has been overstated. Such critiques highlight a gap between habit theory and the reality of human behaviour 'in the wild'. This state-of-the-field review aims to narrow this gap. Building on a core distinction between *habit* and *habitual behaviour*, our review seeks to offer interpretations of habit theory and evidence that will better manage intervention designers' expectations regarding how modifying habit can realistically be expected to promote behaviour change. We emphasise that habit is just one potential influence on behaviour at any given moment, and highlight instances in which habit may dominate over intention, and in which intention may

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dominate over habit, in determining behaviour frequency. We suggest that, while it may assist behaviour maintenance, habit formation may be neither necessary nor sufficient to sustain real-world behaviour change. We draw attention to the various ways in which habit may be 'broken' (i.e., disrupted), and discern the implications of each habit disruption mechanism for long-term cessation of unwanted behaviours.

**KEYWORDS**
automaticity, behaviour, behaviour change, habit, intention, motivation, social psychology

## 1 | INTRODUCTION

Many behaviour change interventions have short-lived success, which erodes as people return to well-established patterns of behaviour (Dombrowski et al., 2010). Making behaviour 'habitual'—that is, creating cue-behaviour associations that generate action automatically on exposure to environmental cues (Wood & Rünger, 2016)—is seen as key to maintaining desirable behaviours over time (Kwasnicka et al., 2016; Rothman et al., 2015). Conversely, breaking 'bad' habits is depicted as central to cessation of longstanding unwanted behaviours (Cleo et al., 2017). Habit-based interventions have shown promise for encouraging health-conducive behaviours and discouraging health-risk behaviours (Cleo et al., 2017, 2020). Habit theory is being adopted to understand and change a growing range of behaviours, spanning sustainability (Linder et al., 2022), cybersecurity (Weickert et al., 2023), and financial decision-making (Sinnewe & Nicholson, 2023).

The habit-based approach to behaviour change has, however, attracted criticism for seemingly failing to realise its potential (Segar, 2022). The persistence of habitual responses even when they no longer serve valued goals, a key characteristic of habit commonly observed among animals in lab-based studies, does not reliably replicate to humans in real-world contexts (de Wit et al., 2018; Gera et al., 2023; Linnebank et al., 2018). Commentators have questioned whether complex behaviours can become fully automated (Maddux, 1997). Contrary to claims that habit formation will extend the longevity of behaviour change, some research suggests habit gains dissipate over time (Judah et al., 2013; Ma et al., 2023). While some have called for the abandonment of habit as a behaviour change mechanism (Segar, 2022), its ubiquitous role in everyday action cautions against such a conclusion.

Critiques of the applicability of habit to real-life settings reveal important gaps between how habit theory is interpreted and experiences of habit 'in the wild'. This state-of-the-field review seeks to narrow these gaps. Given enduring interest in habit and its implications for behaviour change among applied psychologists, practitioners and the general public (e.g., Clear, 2018; Duhigg, 2013; Wood, 2019), it is important that habit theory and evidence is communicated coherently and in a way that is most practically useful. The primary aim of this review is to offer interpretations of habit theory and evidence that will manage expectations about what interventions designed to 'make' and 'break' habits will likely achieve. Our secondary aim is to thereby resolve concerns regarding the applicability of habit to real-world behaviour change settings.

## 2 | HABIT AND HABITUAL BEHAVIOUR

### 2.1 | 'Habit' is not 'habitual behaviour'

There is no 'correct' definition of habit, but definitions that are clear, consistent with theory and evidence, and practical are most useful. Psychologists agree that habitual behaviours arise automatically, through a process whereby exposure to a cue triggers a learnt cue-behaviour association, which in turn generates a behavioural response (e.g., Miller et al., 2019; Verplanken & Orbell, 2022; Wood & Rünger, 2016). Yet, 'habit' itself has been defined variously as the behaviour generated by this process, the automaticity with which behaviour is generated, a tendency to act, or the cue-behaviour association (Fleetwood, 2021; Gardner, 2015; Handel & Smith, 2024; Mazar & Wood, 2018). Some definitions lack coherence in some settings. For example, defining habit as a *type* of behaviour is logically inconsistent with social cognition theories that portray habit as a *determinant* of behaviour (e.g., Hall & Fong, 2007), and portraying habit as a tendency generates the truistic 'prediction' that people with a tendency to act will tend to act.[1]

We prefer to define *habit* as *a cognitive process whereby a contextual cue automatically triggers an impulse to act, based on a cue-behaviour association learnt through consistent repetition in the presence of the cue* (Gardner, 2015). This definition proposes that habit does not generate behaviour directly, but rather activates impulses that energise behaviour (for critiques, see Fleetwood, 2021; Orbell & Verplanken, 2015). In accordance with PRIME Theory (Michie & West, 2013), we view impulses as 'representations of action that … if left uncontested, trigger the specific [neural] pathways that perform the muscle contractions [required to act]' (West & Brown, 2013, p. 197). According to this view, a person with a habit for checking their phone on receiving a notification will automatically reach for the phone when notified (Dowdell & Clayton, 2019), apparently without cognitive mediation; the notification cue will seemingly directly translate into a habitual behavioural response, because the habit impulse is unopposed. PRIME Theory posits that frustrating an impulse arouses a conscious desire to act (West & Brown, 2013); a habitual phone-checker attempting to resist checking their phone after a notification will likely experience an increasingly powerful, conscious urge to do so.

The inclusion of impulses in a habit definition is controversial. Some concerns focus on terminology, because the term 'impulse' risks confusing habitual with impulsive behaviour (Orbell & Verplanken, 2015). To clarify, impulsive behaviour is undertaken with little conscious thought or reflection on possible consequences (American Psychological Association, 2018). An impulse, as we define it, captures the brain activity that energises, coordinates and directs a behavioural response (West & Michie, 2020). Other objections focus on parsimony; habitual action is adequately explained as arising directly from a cued representation of behaviour, rendering impulses a redundant concept (Mazar & Wood, 2018; Orbell & Verplanken, 2015). However, we prefer to conceptually separate the cue-behaviour association from the energising force this association generates, for practical reasons. As we will show later, this separation allows a clearer distinction between habit disruption methods that seek to modify the potential for associations to activate action representations, and those that focus on blocking the translation of such representations into behaviour.

PRIME Theory proposes that all psychological influences on action operate through impulses, and that multiple impulses compete, with behaviour regulated by the strongest momentary impulse (West & Michie, 2020). Specifying the output of the habit process as an impulse, rather than as action itself, serves as a reminder that habit is only one source of impulses, and that competing impulses generated by other psychological processes may dominate. Upon detecting a notification, for example, the behaviour of a habitual phone-checker attempting to reduce phone use will depend on the momentary strength of the intention-driven impulse to avoid phone-checking versus the habit-generated phone-checking impulse. Habit generates behaviour only when there are no stronger, competing impulses (de Wit & Dickinson, 2009).

Our chosen definition distinguishes habit, as a cognitive construct that generates behaviour, from *habitual behaviour*, which refers to *action generated by the habit process*[2] (Figure 1; Gardner & Lally, 2023; de Wit &

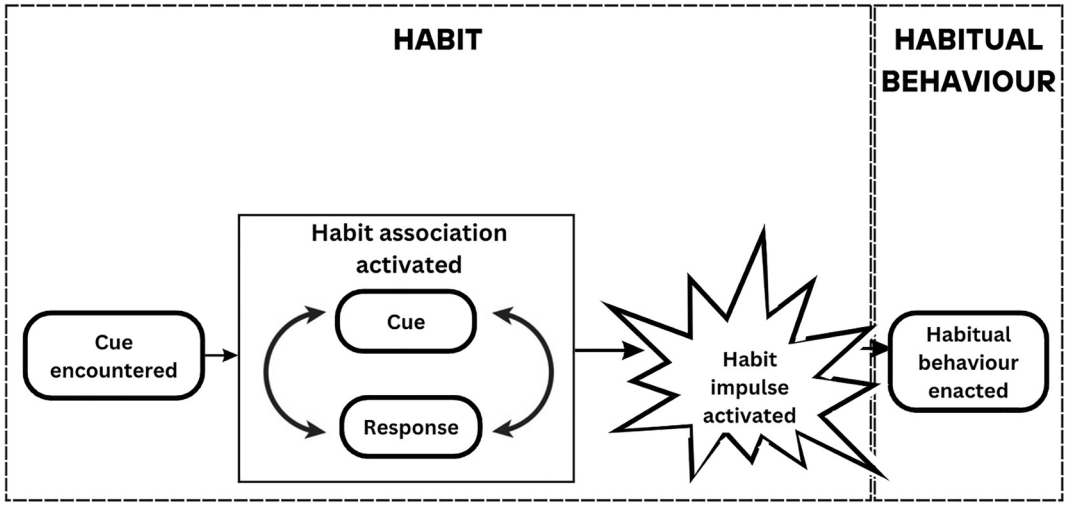



**FIGURE 1** Habit and habitual behaviour: How habit translates into behaviour in the absence of stronger competing impulses.

Dickinson, 2009). As we will argue, this distinction is critical for understanding the profits and pitfalls of habit-based approaches to behaviour change.

## 2.2 | Multiple habits can contribute to any given action

Understanding how habit may support behaviour change requires understanding how habits can manifest in behaviour. Consider the example of 'taking a shower', a commonly habitual action with sustainability implications (Pereira-Doel et al., 2024). A person may have a habit such that, upon waking, they take a shower, which involves a sequence of subservient actions including turning on hot water, soaking the body, lathering up, and so on. This sequence may be underpinned by multiple habit associations. The 'waking' cue initiates the 'taking a shower' sequence. A series of associations, 'chunked' together in memory through repeated sequential performance (Botvinick & Plaut, 2004; Miller et al., 2019), enables progression through the subsequent sequence, such that cessation of one subservient action (e.g. 'turn on hot water') cues the next ('soaking body'). Some commentators have proposed a distinction between habits that trigger (i.e., 'habitually instigate') an episode of behaviour, and habits that automate performance (i.e., 'habitually execute' the action; Gardner et al., 2016). This distinction can be depicted sequentially; adopting an action phases approach (e.g., Heckhausen, 2007), 'instigation habits' effectively bridge the gap between pre-action and action, by selecting and generating a commitment to 'taking a shower', whereas 'execution habits' allow smooth movement through the action phase to completion (e.g., Labrecque et al., 2024). Alternatively, the instigation-execution distinction can be portrayed hierarchically, with 'instigation habits' initiating a higher-order action, within which is nested a series of lower-order actions facilitated by 'execution habits' (Figure 2).

Theorists typically define habitual action as that which is both habitually instigated and executed (Miller et al., 2019; but see Dezfouli & Balleine, 2013). We prefer to use the term 'habitual behaviour' to denote action that is habitually instigated *or* habitually executed or both. Note, however, that the distinction between 'instigation habits' and 'execution habits' is practical, not theoretical; both refer to the same underlying psychological mechanism, but they have different implications for changing a given behaviour. Specifically, associations that habitually

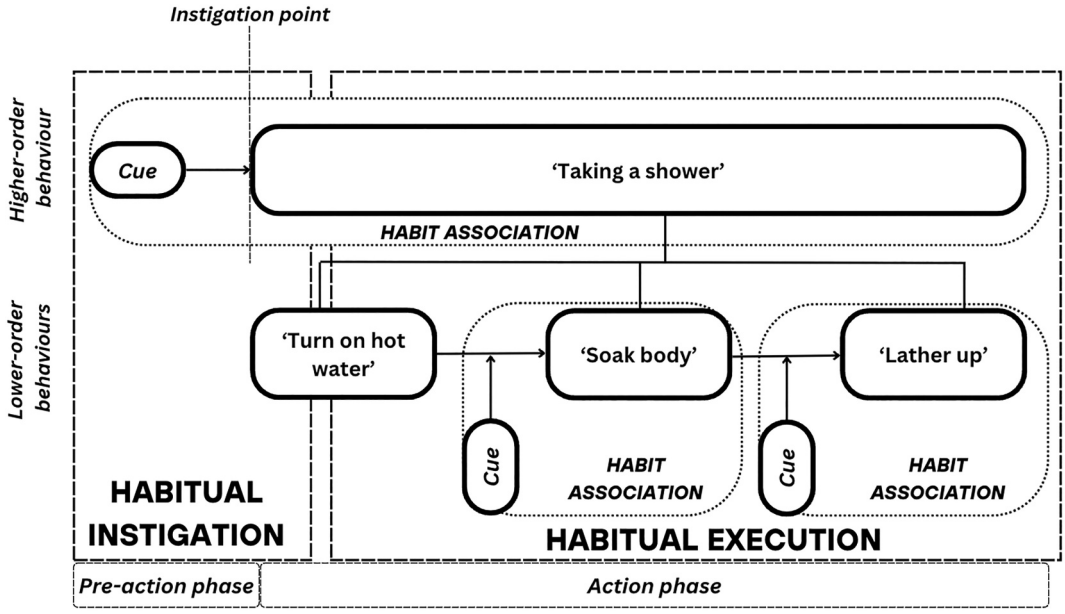



**FIGURE 2** Habitual instigation and execution as applied to showering.

instigate action affect *whether* a behaviour is enacted, whereas associations between subservient behaviours affect *how* the behaviour is performed (Gardner, 2022; Gardner et al., 2016). For showering reduction, disrupting instigation habits would likely decrease showering *frequency*, whereas modifying execution habits might reduce the *duration* of showering episodes.

## 2.3 | Complex behaviours can be habitually triggered

Critics have questioned whether habit associations can support more complex activities, such as socialising (Segar, 2022). An implicit assumption of this criticism is that a behaviour can *either* be habitual *or* intentional, and that effective habit formation interventions will automate behaviour to the extent that intentional (i.e., reflective) processes are no longer required. Yet, this is a false dichotomy. While simple behaviours for which initiation and execution occur rapidly and in close succession, such as flicking a switch, may be entirely automated, more intricate behaviours are often driven by a combination of automatic and reflective processes (Phillips & Mullan, 2023; Saunders & More, 2024); thus, *a behaviour may be at once both habitual and intentional*. For example, a person who consciously decides to take a shower (i.e., intentional instigation) may enact the in-shower sequence automatically (habitual execution), or a person automatically triggered to shower (habitual instigation) may mindfully undertake in-shower activities (intentional execution).

Recognising the operation of habits in different action phases can resolve debates around the relevance of habit to complex actions (Gardner & Lally, 2023; Phillips & Mullan, 2023; Rebar et al., 2023). Behavioural complexity refers to the intricacy of *executing* an action, not *instigating* it. While complex behaviours may take longer to become habitually triggered (Buyalskaya et al., 2023; Lally et al., 2010), even actions requiring complex performance can be habitually instigated (Gardner, 2022). Habit formation approaches need not seek to *fully* automate complex behaviours; forming instigation habits should promote frequent repetition of simple and complex actions alike (Gardner, 2022; but see Phillips & Mullan, 2023).

## 3 | THE ROLE OF HABIT AND INTENTION IN BEHAVIOURAL FREQUENCY

Social psychology theory surrounding habit as a determinant of action, and therefore a behaviour change mechanism, has implicitly emphasised habitual *instigation* and action *selection* as precursors of action frequency (Gardner et al., 2016). From hereon we focus solely on 'instigation habits' and action selection.

A theorised hallmark of habit is its potential to override intentions as the predominant influence on action (Dickinson, 1985; Triandis, 1977; Wood et al., 2022): as habit associations form, they acquire the potential to generate action even when the original goal served by the action loses its value (Wood et al., 2022; but see de Houwer et al., 2023). Much evidence suggests that, as habit strengthens, the likelihood that intention will inform action in the presence of habit cues decreases (Gardner et al., 2011; Rebar et al., 2016). This 'habit-intention interaction hypothesis' underpins interest in using habit to promote long-term behaviour change: it is proposed that making 'good' habits will sustain behaviour even if conscious motivation declines, while breaking 'bad' habits is necessary to prevent relapse into long-standing unwanted action (Verplanken & Wood, 2006). Yet, research into the long-term impact of habit change on behaviour maintenance is scarce (Gardner & Rebar, 2019), and one study suggested that habit change may not translate into long-term behaviour (Judah et al., 2013). The hypothesised dominance of habit over intention warrants closer examination.

### 3.1 | Habit and intention can complementarily determine action selection

The considerable research attention paid to instances in which habit and intention compete might mislead practitioners into viewing habit and intention as inherently conflicting influences. Importantly, habit and intention often correspond and can combine to promote action selection. Habit forms through consistent in-context repetition (e.g., Keller et al., 2021; Kilb & Labudek, 2022), which, assuming free will, is likely driven by strong, enduring intentions (Lally & Gardner, 2013). Habit often represents a 'locked in' response that, while no longer *triggered* by intention, is nonetheless *aligned* with intention (Wood et al., 2022).

When habit and intention concur, habit can *strengthen* the translation of intention into action. For example, people who intend to use sunscreen are more likely to act if they have strong sunscreen use habits (Rebar et al., 2022). This suggests that habit will ensure sunscreen use among intenders even when positive intention is momentarily weakened, or they might otherwise forget to act (Neal et al., 2013). People often form habits as a purposeful self-regulatory strategy, to maintain self-discipline amid momentary temptations (Adriaanse et al., 2014; Brinkhof et al., 2023; de Ridder & Gillebaart, 2017; Lin et al., 2016). People may also deliberately expose themselves to contexts associated with their habits, to encourage goal-supportive behaviour (Williamson & Wilkowski, 2022); a person with a habit for exercising in the gym, for example, may intentionally attend the gym to trigger their exercise habit despite lacking momentary enthusiasm.

### 3.2 | Strong and stable counterhabitual intention may override habit in the long-term

Studies suggesting that habit overpowers *counterhabitual* intention in determining action selection often lack ecological validity. Real-world tests have typically pitted strong habit against weakened habit-aligned intention (e.g., spending habit, weak spending intention), but a weak intention towards one action (e.g., spending) is not a strong intention towards a competing action (e.g., avoid spending; Gardner, Lally, & Rebar, 2020). Studies of true conflict, pitting habit against counterhabitual intention (e.g., spending habit, intention to avoid spending), have found mixed results. Lab-based studies suggest habitual behaviours can persist despite conflicting intentions (Webb et al., 2009), yet observational studies often fail to find such effects (Gardner, Lally, & Rebar, 2020).

The potential for habit to override counterhabitual intention in lab settings may fail to replicate in real-life contexts for several reasons. Lab tasks are often explicitly designed to restrict participants' self-regulatory capacity, by for example, curtailing response times (Hardwick et al., 2019; Verplanken et al., 1994). Such tasks generate 'pure' illustrations of habit-driven responses but lack generalisability to real-world settings in which people have sufficient self-regulation to frustrate unwanted habits (Quinn et al., 2010). Habit is a stronger predictor of behaviour when people lack self-control (e.g., when distracted or fatigued) (Mylopoulos, 2022; Neal et al., 2013; Pfeffer & Strobach, 2022), but where people have the capacity to inhibit unwanted habit tendencies, counterhabitual intention is a stronger predictor of action than is habit (Conner et al., 2023).

Lab-based habit tasks also typically focus on simple behaviours for which opportunities to implement self-control are limited. For actions such as pressing computer keys, for example, acting counterhabitually requires self-regulation immediately prior to the key-press. Complex real-world behaviours usually offer greater opportunity for inhibition. Habitual smokers, for example, might exercise self-control by inhibiting initiation of an action sequence (e.g., before reaching for a cigarette), or aborting progression *after* the sequence is initiated (e.g., after reaching for the cigarette but before lighting it; Orbell & Verplanken, 2010). Strong unwanted habits are less likely to translate into overt action in real-world settings because people have greater opportunity to align their behaviour with counterhabitual intentions than they do in the lab (Wood et al., 2022).

Although lab-based studies show that people can make 'action slips' when attempting to act counterhabitually (e.g., de Wit et al., 2012), habit-induced slips are, by definition, exceptional failures to act as intended (Mylopoulos, 2022). Despite occasional slips, over time people adjust their action and discontinue the habitual behaviour, even in simplistic lab tasks (Bouton, 2021). Although people often fall back into well-established unwanted behaviours following early behaviour change successes (Dombrowski et al., 2010), full relapse is not solely underpinned by 'bad' habits per se, but rather attributable to a complex cascade of negative cognitive and emotional responses *sparked by* a habit-based action slip (Ten Broeke & Adriaanse, 2023). For example, failure to avoid habitual snacking on one occasion may be interpreted by a person with clear dietary goals as an irredeemable violation of rules, causing them to perceive maintenance as worthless, prompting further self-regulation failure and unravelling behaviour change (Ten Broeke & Adriaanse, 2023).

While habit may overpower intention *on occasion*, there is no compelling evidence to show that habit will *persistently* overpower intention in determining action frequency, *across multiple instances*, over the long-term. Drawing on Gardner, Lally, and Rebar (2020), Figures 3a,b offer an alternative hypothesised interaction between a 'bad' habit—that is, taking a bath, as opposed to the more water-conserving activity of showering—and intention as predictors of behaviour, in the absence and presence of self-control, respectively. The example assumes that taking a bath or shower are the only options and are diametrically opposed, and that environmental contexts remain stable. Where a person lacks self-control, the frequency of taking a bath will be determined by habit strength alone,

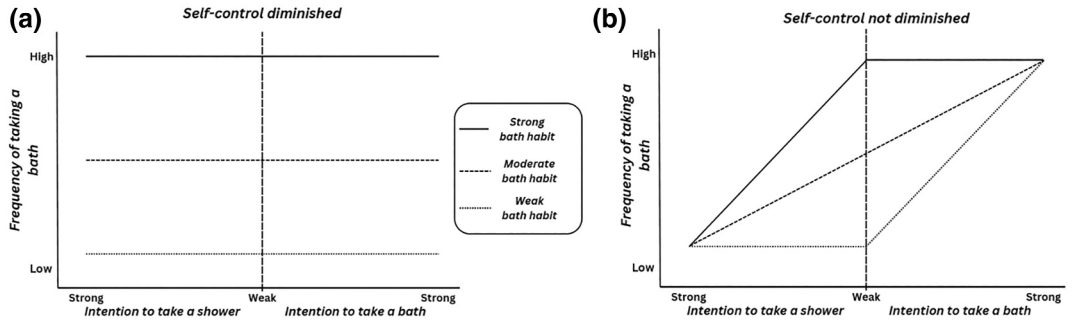



**FIGURE 3** Hypothesised interaction between habits and intentions, at differing levels of self-control, as applied to habitual bathing.

regardless of intention (Figure 3a). Where a person maintains sufficient self-control (Figure 3b), habit-intention interplay depends on the direction of intention. Bath-taking habit will tend to override weakened intention to conserve water in determining frequency, but strong counterhabitual intention to conserve water will tend to override even strong bath-taking habit. The figure predicts that, in unvarying contexts in which a person has sufficient motivation and self-regulatory capacity, intention will generally overpower habit.

## 4 | HOW HABIT FORMATION CAN—AND CANNOT—PROMOTE LONG-TERM BEHAVIOUR CHANGE

### 4.1 | Habit formation may be neither necessary nor sufficient to sustain long-term behaviour change

Claims that habit is key to sustaining behaviour change over time must be qualified (Rothman et al., 2015). First, habitual behaviours are cue-dependent (Orbell & Verplanken, 2010); environmental changes that disrupt cue exposure will discontinue habitual action (see Verplanken et al., 2018). Second, any newly adopted behaviour can be sustained by non-habit mechanisms, such as enhanced liking, or satisfaction with the decision to change (Baldwin et al., 2006; Volpp & Loewenstein, 2020). While habit is conducive to behaviour change, there is no compelling evidence to our knowledge that habit formation is *essential* for long-term behaviour maintenance.[3] Although many people who successfully change their behaviour develop habits (e.g., Kwasnicka et al., 2023), behaviour maintenance cannot reliably be interpreted as evidence of habit formation (Volpp & Loewenstein, 2020).

As Figure 3b indicates, even when habit has formed, habit impulses may be overpowered by strong counterhabitual intentions. This undermines the assumption that habit will sustain behaviour change regardless of motivation. Making a wanted behaviour habitual should protect against some erosion of positive motivation, but if motivation changes such that a person wishes *not* to do the behaviour, or strongly intends to perform an alternative, a habitual behaviour may be discontinued over time. Initially successful habit formation attempts may therefore fail to sustain behaviour because people choose to cease newly adopted behaviours (Judah et al., 2013). Habit formation may not therefore be sufficient to sustain behaviour; even for habitual behaviours, long-term maintenance will likely require a base level of positive motivation towards the habitual behaviour, or at least an absence of motivation to *cease* performance (Gardner, Lally, & Rebar, 2020). Interventions might fruitfully not only encourage development of positive habits (see Gardner, Rebar, & Lally, 2020), but also sustain motivation to support habitual behaviour.

### 4.2 | Habit formation creates memory traces that may (re)activate behaviour when desired

Although a *habitual behaviour* may cease over time if people become motivated to abandon it, or cue exposure ceases, the underlying *habit association* may persist in memory. Theory and evidence may appear inconsistent regarding the persistence of habit associations after discontinuation of behaviour. Some commentators argue that, when a habitual behaviour is discontinued, habit associations naturally weaken (Walker et al., 2015), while others argue that habit associations cannot be unlearnt (Bouton, 2000). These viewpoints may be reconciled by proposing that unenacted habits can be weakened but not fully extinguished, so retain capacity to activate habitual behaviours when cued. This explains why old habitual behaviours are recovered after periods of dormancy. For example, following temporary road closures that prohibit car use, some drivers quickly return to driving habits (Brown et al., 2003; see too Ratay et al., 2024). In the longer-term, people who successfully modify their diet often report ongoing struggles to avoid lapses in familiar contexts, months after initiating change (Gardner et al., 2021).

Commentators have largely overlooked the possible benefits of forming 'good' habit associations that outlast habitual behaviour. Forming a habit may not sustain a behaviour that has lost all motivational value but should in theory allow for rapid and relatively effortless recovery of that behaviour in associated settings if it regains its value. For example, we speculate that people who developed mask-wearing habits during the COVID-19 pandemic should be better placed to recover mask-wearing in future pandemics (Li et al., 2023). Further research is needed to test this prediction.

## 5 | HOW HABIT DISRUPTION CAN—AND CANNOT—CHANGE LONG-TERM BEHAVIOUR

Mapping the process through which habitual behaviour is generated reveals distinct practical strategies to disrupt (or 'break') habitual responses (Figure 4). There is no universal 'best' strategy; the utility of each strategy will vary according to the actor, environment and target behaviour.

### 5.1 | Disrupting habitual behaviour

*Habit inhibition* involves wilful frustration of a habit impulse. This may entail stopping oneself from acting prior to initiating a behaviour (e.g., thinking 'don't do it!' when cued to gamble online) or abandoning an ongoing action sequence prior to completion (e.g., closing a casino website before depositing money). While participants in one study rated this strategy particularly effective (Quinn et al., 2010), habit inhibition requires insight into contextual cues, and the momentary motivation and willpower to vigilantly monitor cues and inhibit the unwanted response, which can be cognitively effortful.

*Reducing behavioural accessibility* and *habit cue discontinuity* respectively limit availability of a habitual behaviour after the impulse is triggered or diminish the likelihood that a habit impulse will be activated, and may be temporary or permanent, purposeful or incidental.[4] Reducing accessibility involves 'adding friction' such that an action

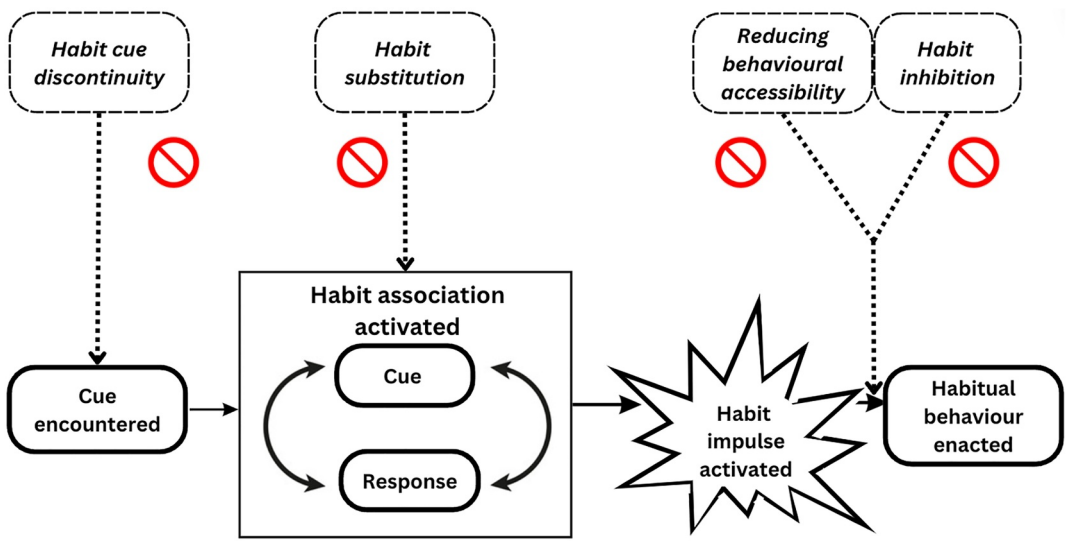



FIGURE 4 Habit disruption mechanisms, mapped to the process through which habit translates into habitual behaviour.

becomes wholly unavailable or otherwise prohibitively effortful to enact (Cheval & Boisgontier, 2021; Wood & Neal, 2016). Purposeful accessibility reduction may for example, take the form of intentionally blocking gambling transactions on payment cards so that online gambling cannot easily be enacted, while incidental reduction might arise from casino payment servers failing, precluding gambling. Purposefully reducing behavioural accessibility requires the capability and opportunity to modify access in advance of the habit impulse being activated; an individual unable to remember their banking credentials may not be able to block their payment card, for example, so will be less able to prevent habitual gambling.

Purposeful habit cue discontinuity involves knowingly limiting exposure to triggers to unwanted responses (i.e., stimulus control), such as avoiding socialising with friends that gamble online (Schüz et al., 2017). Purposeful cue discontinuity requires awareness of potential cues, and that cues can feasibly be avoided; it is unrealistic, for example, to expect a person habitually triggered to gamble by negative emotional states to avoid negative affect altogether. Incidental discontinuity, on the other hand, arises from naturally occurring context changes that diminish exposure to cues; becoming a parent, for example, may limit exposure to online gambling cues. Incidental reduced accessibility and cue discontinuity have immediate impact on habitual behaviour but are by definition outside of the control of the actor (but see Williamson & Wilkowski, 2022).

While reduced accessibility and cue discontinuity can offer potential 'windows of opportunity' for deviating from old patterns and developing new, wanted habits (Verplanken et al., 2018), temporary inaccessibility or cue discontinuity may prompt only short-lived disruption of habitual behaviours (e.g., Tohidi et al., 2022). Permanent removal of habitual options or lasting context change should achieve durable disruption (Verplanken et al., 2018). However, inhibiting habitual responses, reducing access or avoiding cues may not extinguish habit *associations* (Gardner et al., 2021). Persistent habit associations may reactivate unwanted behaviour when the willpower needed to inhibit habitual responding is diminished, unwanted habitual behaviours become easily accessible, or old habit cues are re-encountered.

### 5.2 | Disrupting habit

*Habit substitution* refers to the direct displacement of one habit association with another. Consistently performing an alternative behaviour in response to a cue to an existing, unwanted habitual behaviour should engender a new, stronger and dominant association that produces a new default response. In practice, this requires elements of habit inhibition or reducing behavioural accessibility to block an unwanted habit, and habit formation strategies to encourage a wanted alternative. For example, habitual snackers may be encouraged to avoid buying unhealthy snacks (i.e., reducing behavioural accessibility) and repeatedly choose a healthier option (habit formation), to form new healthier snacking habits (Juszczyk & Gillison, 2018). Habit substitution directly targets underlying habit associations, so should minimise future lapses into unwanted habits (Wood & Neal, 2016). In practice however, substitution may be undermined if the new behaviour is less appealing than the habitual action (Baldwin et al., 2006), or the actor struggles to consistently obstruct the old response.

## 6 | IMPLICATIONS FOR HABIT-BASED INTERVENTIONS

### 6.1 | Implications for habit formation interventions

'Forming a habit' does not mean creating a new, settled pattern of behaviour, but rather involves learning cue-behaviour associations that, when cued, automatically generate action impulses. However, habit impulses do not inevitably generate behaviour. Habit formation interventions can automate wanted actions and protect against dips in motivation, but an actor with sufficient momentary self-control will likely discontinue performance of unwanted

habitual actions over time even if a habit impulse is activated. Habit formation interventions should be augmented with strategies to sustain favourable motivation.

Intervention designers should consider whether a habit formation approach is appropriate in a target setting. Where habit formation is unfeasible—for example, among people with unpredictable or unstructured everyday lives (van de Vijver, et al., 2023)—various non-habit mechanisms may better support lasting behaviour change (Volpp & Loewenstein, 2020). In settings in which context-consistent repetition of a wanted action is feasible, habit formation should be considered as an intervention approach. For simple and complex behaviours alike, developing a habit should sustain engagement in behaviour, assuming lasting positive motivation. Forming a habit for instigating a behaviour (e.g., 'eating vegetables'), but permitting more reflective execution via various, more specific behaviours (e.g., 'eating carrots') may offer an attractive blend of automaticity and flexibility (Hagger, 2019; Phillips et al., 2019).

## 6.2 | Implications for habit disruption interventions

Habit disruption approaches should be used only where there is reason to believe a behaviour is habitual, as opposed to merely long-established. Employing habit disruption strategies will not necessarily permanently discontinue habitual behaviour. Strategies based on reducing behavioural accessibility or cue exposure may not always be feasible. Inhibition-based strategies depend on sufficient self-regulation to frustrate responses on multiple occasions, so may be unsuitable for people with long-term self-regulatory impairments (van de Vijver et al., 2023). Additionally, if momentary self-control cannot be maintained, an unwanted behaviour becomes more accessible, or cue discontinuation is temporary, a persistent habit association may prompt lapses into long-inactive habitual behaviours, thereby triggering cognitions and emotions that fatally undermine behaviour change (Ten Broeke & Adriaanse, 2023). Habit-disruption interventions should frame lapses as temporary, permissible setbacks, not irreversible failures.

By focusing on cue-behaviour associations that generate habitual responding, habit substitution should have enduring impact on behaviour. Habit substitution intervention designers must consider whether, when, and how to employ inhibition or behavioural inaccessibility strategies to block established habitual behaviour most effectively, and how best to promote the context-consistent repetition necessary to form a new, alternative habit.

Disruption strategies seek to nullify the influence of old habit associations on momentary impulses, freeing people to act in line with conscious motives. However, habit disruption strategies will only meaningfully change behaviour among people who remain motivated and capable of acting contrary to their 'bad' habits at critical moments. A person who initially inhibits habitual nightly social media use, for example, will likely lapse if they continue to derive enjoyment from this behaviour (Hill et al., 2024).

# 7 | CONCLUSION

Common interpretations of habit theory often conflate habit, as a process that generates behaviour, and habitual behaviour generated by this process. This has led intervention developers to overestimate the utility of habit as a behaviour change mechanism, prompting criticisms about the apparent failure of habit-based approaches to achieve desired outcomes (Segar, 2022). The principles that we have outlined, regarding habit theory as it applies to the real-world, should better manage intervention developers' expectations regarding what successful habit formation or disruption interventions can realistically achieve.

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## CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflicts of interest.

## DATA AVAILABILITY STATEMENT

No data were generated as part of this research.

## ETHICS STATEMENT

Not applicable.

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## ENDNOTES

[1] While we caution against *defining* habit as a tendency, we recognise that there may be individual differences in the extent to which people *tend to form* habits or *tend to act* on learnt cue-response associations (Ersche et al., 2017; Verhoeven & de Wit, 2018).

[2] Other theorists define habit as the cue-behaviour association that is learnt through cue-consistent performance (Fleetwood, 2021; Phillips & Mullan, 2023). We prefer to term this a 'habit association', which is one component of the broader process through which cued associations activate habitual action in relevant contexts. Definitions of habit as a process or as the association at the heart of this process are however equally satisfactory for the purpose of conceptually distinguishing habit and habitual behaviour and have identical practical implications.

[3] It is, however, plausible that individual differences may shape the centrality of habit formation to behaviour maintenance such that, for some people, habit may be more implicated in sustaining behaviour than it is in others (Verhoeven & de Wit, 2018).

[4] Incidental habit cue discontinuity has been labelled elsewhere as 'habit discontinuity' (Verplanken et al., 2018; Walker et al., 2015). We prefer the term 'habit cue discontinuity' for its greater precision and explanatory value, as it denotes the mechanism through which habitual behaviour is discontinued (i.e., reduced cue exposure).

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